



# 9 – Population and CLT

*Tutor: Chin Yi*

# Recap: Sample

## Example 4

Let  $X_1, \dots, X_n$  be a random sample.

- **Sample mean:**  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
- **Sample variance:**  $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ .

Note that  $\bar{X}$  and  $S^2$  are statistics. Their distributions are examples of sampling distributions.

$$\mathbb{E}(S^2) = \frac{1}{n-1} \sum_{i=1}^n \mathbb{E}((X_i - \bar{X})^2) = \sigma^2. \quad \text{*Proof in further readings}$$

# Recap: Sample Normalization

## Theorem 5 (Random sample from Normal distribution)

Let  $X_1, \dots, X_n$  be observations of a random sample of size  $n$  from the normal distribution  $N(\mu, \sigma^2)$ . Then the sample mean  $\bar{X}$  and sample variance  $S^2$  are independent, and

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} \sim \chi^2(n-1).$$

## Theorem 6 (Random sample from Normal distribution)

Let  $X_1, \dots, X_n$  i.i.d  $\sim N(\mu, \sigma^2)$ . The sampling distribution of the sample mean  $\bar{X}$  is given by

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

This implies that the standardized sample mean  $\frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}} \sim N(0, 1)$ .

How to solve:

Suppose  $Y = f(X_1, X_2, \dots, X_n)$

We find:

1.  $E(Y)$
2.  $\text{Var}(Y)$

Then we use the formula as such

$$\frac{Y - E(Y)}{\sqrt{\text{Var}(Y)}} \sim N(0, 1)$$

We use the  $\text{sqrt}(\text{Var}(Y))$  as the denominator

# Recap: CLT

## Theorem 8 (Central Limit Theorem (CLT))

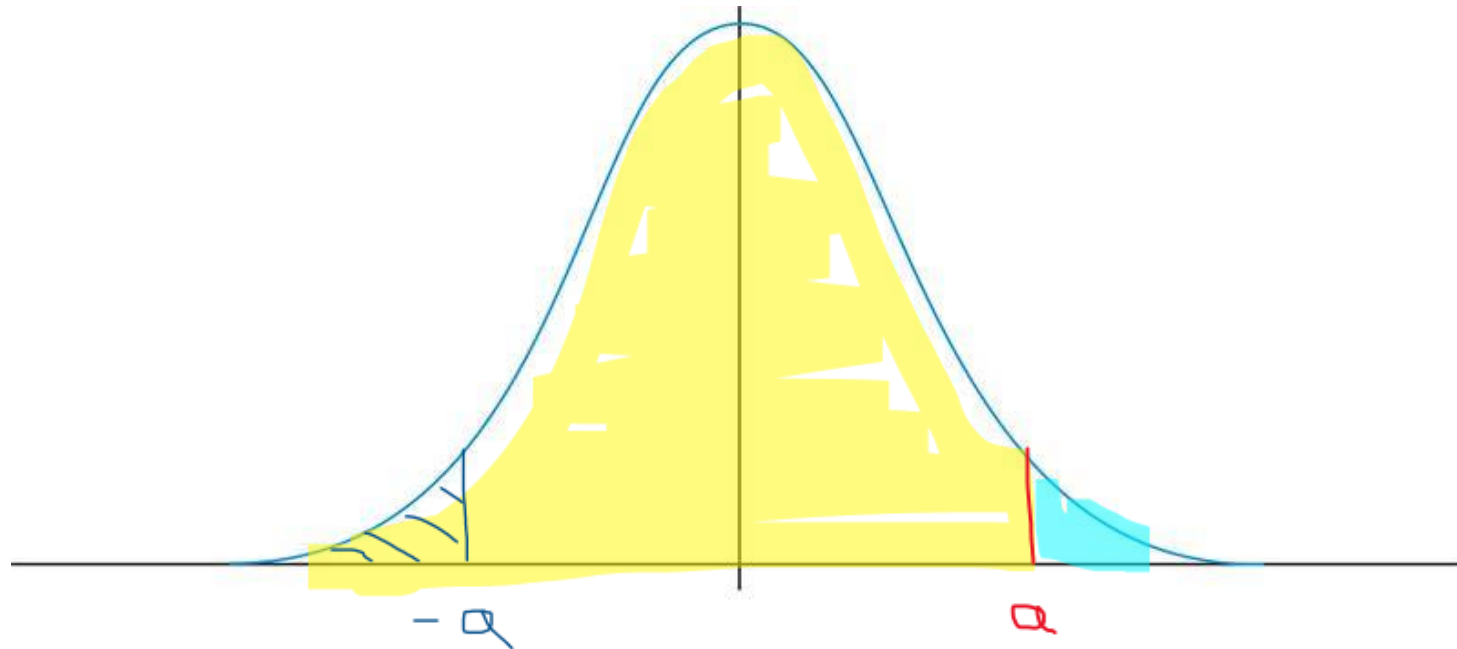
*Let  $X_1, \dots, X_n$  i.i.d with  $\mathbb{E}[X_i] = \mu$  and  $\text{Var}[X_i] = \sigma^2 < \infty$ . Then*

$$\mathbb{P} \left( \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq x \right) \rightarrow \Phi(x) \text{ for } n \rightarrow \infty.$$

*Here,  $\Phi(x)$  is the CDF of standard normal.*

# Recap: CLT Symmetry

$$\Phi(a) = 1 - \Phi(-a) \text{ for all } a > 0.$$



The Normal Distribution

### Problem 1 (Distribution of Sample Mean)

- (a) Let  $X_1, \dots, X_{100} \sim N(0, 1)$ . Find  $P(|\bar{X}| > 0.1)$  where  $\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$ .
- (b) Let  $X_1, \dots, X_n$  be i.i.d  $\sim \text{Gamma}(\alpha, \theta)$ . Use MGFs to determine the distribution of  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ .  
MGF of  $\text{Gamma}(\alpha, \theta)$  is  $(1 - \theta t)^{-\alpha}$ .
- (c) Let  $X_1, \dots, X_{100}$  be i.i.d  $\sim \text{Gamma}(10, 10)$ . Use CLT to approximate  $\mathbb{P}(95 \leq \bar{X} \leq 105)$ .
- (d) Let  $X_1, \dots, X_n$  be an i.i.d sample drawn from a population distribution  $D$ . Is there any relation between the sample mean and the population mean?

(a)  $\mathbb{P}(|\bar{X}| > 0.1) \approx 1 - (0.841 - 0.159) = 0.318.$

(c) 
$$\begin{aligned} \mathbb{P}(95 \leq \bar{X} \leq 105) &= \mathbb{P}(95 \times 100 \leq \sum_{i=1}^{100} X_i \leq 105 \times 100) \\ &\approx \Phi\left(\frac{105 \times 100 - 100 \times 100}{\sqrt{1000}\sqrt{100}}\right) - \Phi\left(\frac{95 \times 100 - 100 \times 100}{\sqrt{1000}\sqrt{100}}\right) \\ &= \Phi(1.5811) - \Phi(-1.5811) \\ &= 0.9429 - 0.0571 = 0.8858. \end{aligned}$$

$$\frac{Y - E(Y)}{\sqrt{Var(Y)}} \sim N(0, 1)$$

**Problem 2 (Distribution of Sample Mean)**

Let  $X_1, \dots, X_n$  be i.i.d  $\sim N(10, 100)$ . How large does  $n$  need to be such that

$$P(|\bar{X} - 10| < 0.001) > 0.99 \quad ?$$

$$\Phi(-10^{-4}\sqrt{n}) = 0.005. \longrightarrow n \approx 6.636 \cdot 10^8. \not\approx$$

| $z_\alpha$ | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0        | 0.5000 | 0.4960 | 0.4920 | 0.4880 | 0.4840 | 0.4801 | 0.4761 | 0.4721 | 0.4681 | 0.4641 |
| 0.1        | 0.4602 | 0.4562 | 0.4522 | 0.4483 | 0.4443 | 0.4404 | 0.4364 | 0.4325 | 0.4286 | 0.4247 |
| 0.2        | 0.4207 | 0.4168 | 0.4129 | 0.4090 | 0.4052 | 0.4013 | 0.3974 | 0.3936 | 0.3897 | 0.3859 |
| 0.3        | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |
| 0.4        | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| 0.5        | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| 0.6        | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| 0.7        | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| 0.8        | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| 0.9        | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| 1.0        | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| 1.1        | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| 1.2        | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| 1.3        | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| 1.4        | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| 1.5        | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| 1.6        | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| 1.7        | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| 1.8        | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| 1.9        | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| 2.0        | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| 2.1        | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| 2.2        | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| 2.3        | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| 2.4        | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| 2.5        | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| 2.6        | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |

$$\frac{Y - E(Y)}{\sqrt{Var(Y)}} \sim N(0, 1)$$

### Problem 3 (Sample Mean and Variances as Estimators)

The strength of chess players is measured by the so-called Elo rating. From a group of 1000 chess players, the Elo ratings of 10 players are sampled with the following results  $x_1, \dots, x_{10}$ .

1628, 1534, 1630, 1634, 1789, 1443, 1584, 1353, 1559, 1430

We assume that the population distribution is  $N(\mu, \sigma^2)$  with unknown  $\mu$  and  $\sigma$ .

- (a) Compute the sample mean  $\bar{x} = \frac{1}{10} \sum_{i=1}^{10} x_i$  and sample variance  $s^2 = \frac{1}{9} \sum_{i=1}^{10} (x_i - \bar{x})^2$  based on the observations  $x_1, \dots, x_{10}$ .
- (b) The sample mean can be used to approximate  $\mu$  and the sample variance to approximate  $\sigma^2$ . Based on this, find the approximate probability that a randomly chosen player from the group of 1000 players has an Elo rating higher than 1700.

(a) We get  $\bar{x} = 1558.4$  and  $s^2 = 15789.6$ .

$$\begin{aligned} \mathbb{P}(X > 1700) &= 1 - \mathbb{P}(X < 1700) \\ &= 1 - \mathbb{P}\left(Z < \frac{1700 - 1558.4}{\sqrt{15789.6}}\right) \\ &\approx 1 - \Phi(1.13) = 1 - 0.8708 \approx 0.13. \end{aligned}$$

#### Problem 4

Let  $X_i$  be the time between the  $i$ th and  $(i+1)$ th eruption of a volcano and assume that  $X_1, \dots, X_n$  are i.i.d  $\sim \text{Exp}(\theta)$ , where  $\theta$  is unknown. The following observations  $x_1, \dots, x_{10}$  (time in years) have been made.

1343.4, 1753.2, 1569.8, 645.4, 2617.0, 3897.0, 348.7, 3017.3, 2197.3, 245.1

Write  $\bar{x} = \frac{1}{10} \sum_{i=1}^{10} x_i$ .

Explain why  $\bar{x}$  is a reasonable estimation of  $\theta$ .

\* We will touch more on this when we reach **Bias** and **Standard Deviation**.

$$\frac{Y - E(Y)}{\sqrt{Var(Y)}} \sim N(0, 1)$$

**Problem 5 (Use of CLT for approximation)**

Let  $Y = X_1 + X_2 + \cdots + X_{15}$  be the sum of i.i.d random sample of size 15 from the distribution whose PDF is

$$f(x) = \frac{3}{2}x^2, \quad -1 < x < 1; \quad f(x) = 0 \quad \text{elsewhere.}$$

Use the central limit theorem to approximate the probability  $\mathbb{P}(-0.3 \leq Y \leq 1.5)$ .

$$\begin{aligned} \mathbb{P}(-0.3 \leq Y \leq 1.5) &= \Phi\left(\frac{1.5 - 15(0)}{\sqrt{0.6}\sqrt{15}}\right) - \Phi\left(\frac{-0.3 - 15(0)}{\sqrt{0.6}\sqrt{15}}\right) = \Phi(0.5) - \Phi(-0.1) \\ &= 0.6915 - 0.4602 = 0.2313. \end{aligned}$$